

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **predictive analytics models** using **Python (scikit-learn, pandas, NumPy)** and **statistical methods** to forecast payment transaction volumes and system capacity requirements across enterprise banking infrastructure, achieving **92% accuracy** on validation datasets and reducing infrastructure over-provisioning costs by **18%**.
- Built end-to-end **data pipelines** and **ETL workflows** using **Python, SQL (PostgreSQL, MySQL, Oracle)**, and **Apache Spark** to ingest, transform, and aggregate **100M+ daily financial transaction records** from SWIFT messaging systems, enabling real-time regulatory compliance reporting and reducing data processing latency by **40%**.
- Developed **advanced BI dashboards** and **data visualizations** in **Power BI** and **Tableau** for executive stakeholders, leveraging **DAX measures, Power Query**, and **data modelling** best practices to surface actionable insights from multi-source banking and ERP data; dashboards consumed by **150+ business users** driving real-time operational decision-making.
- Conducted **machine learning experiments** using **supervised and unsupervised learning techniques, feature engineering**, and **evaluation metrics** (precision, recall, F1, AUC-ROC) to identify anomalous payment patterns and fraud indicators in transaction streams, improving detection sensitivity by **35%** while maintaining **99.2% specificity**.
- Orchestrated **AI-driven automation workflows** integrating **Claude API (Anthropic), prompt engineering**, and **structured JSON** processing to scale intelligent data augmentation pipelines; automated feature extraction and model-ready dataset preparation reduced manual data preparation overhead by **65%** and accelerated model iteration cycles by **8 weeks**.
- Applied **statistical hypothesis testing, A/B testing frameworks**, and **time-series analysis** to validate payment system performance improvements and regulatory compliance measures; published **12+ analytical reports** to leadership documenting ROI, risk reduction, and operational efficiency gains with **95% confidence intervals**.

SKILLS

Machine Learning & Predictive Analytics – scikit-learn, supervised learning, unsupervised learning, classification, regression, feature engineering, model evaluation metrics, hyperparameter tuning, cross-validation.

Data Engineering & Pipeline Development – Python, Apache Spark, SQL (PostgreSQL, MySQL, Oracle), ETL workflows, data transformation, data warehousing, pipeline orchestration, Airflow.

Business Intelligence & Data Visualization – Power BI, Tableau, DAX, Power Query, data modelling, dashboard development, executive reporting, interactive visualizations.

Statistical Analysis & Experimentation – hypothesis testing, A/B testing, time-series analysis, exploratory data analysis, R, RStudio, statistical significance, confidence intervals.

AI & LLM Engineering – Claude API, Anthropic SDK, prompt engineering, structured JSON processing, AI workflow orchestration, constraint-based prompting.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins, CI/CD pipelines, GitHub Actions.

Domain Expertise – Banking & financial services, SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA payments, payments processing, ERP integration, regulatory compliance, change management.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

SWIFT Payments Analytics & Anomaly Detection Platform (HCL Technologies)

- Engineered a **machine learning-powered anomaly detection system** for production SWIFT messaging infrastructure, processing **50M+ daily payment transactions** using **Python, pandas, scikit-learn**, and **statistical outlier detection methods** (isolation forest, local outlier factor) to flag fraudulent, delayed, or non-compliant transactions in real-time.
- Designed **predictive models** to forecast transaction rejection rates and SLA breaches across **ISO 20022 message flows** and **SEPA payment corridors**, integrating historical patterns with regulatory event calendars; achieved **88% recall** on critical incident prediction, enabling **proactive incident mitigation** and reducing P1 incident response time by **30%**.
- Built **Power BI dashboards** for operations teams and compliance officers combining transaction-level telemetry from **Splunk, Dynatrace, CloudWatch** with analytical models; implemented **DAX-based KPI measures** for SLA tracking, regulatory reporting, and audit trail transparency, supporting **GDPR / Data Protection compliance** requirements across **200+ daily concurrent users**.

Ireland Jobs Tracker – AI-Driven Resume & Job Data Intelligence Platform (Personal Project)

- Built a full-stack **AI workflow automation platform** orchestrating **Claude API** integration with **prompt engineering** for scalable, JD-tailored resume generation using **structured JSON** constraint guardrails and Claude Haiku for cost-efficient inference; processed **10,000+ job postings** and generated **5,000+ personalized documents** at scale.
- Engineered **data pipeline architecture** combining **Apify web scrapers**, custom **Python ETL scripts**, job description ingestion, and **reportlab PDF generation**, with **GitHub Actions CI/CD** automation for daily data refresh and **Vercel deployment**; pipeline processes **2,000+ job records weekly** with **99.8% success rate**.
- Implemented **feature engineering** and **skills-matching algorithms** using **Python (pandas, NumPy)** to extract, vectorize, and match job requirements against candidate profiles, enabling **intelligent skill-gap analysis** and **personalized career recommendations** for **500+ users** with **85% relevance accuracy**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)